PASAN UDARA

+94 70 5693282 | pasanudara356@gmail.com | linkedin.com/in/pasan-udara | github.com/PasanUdara

AI/ML-focused Data Science undergraduate with one year of hands-on ML and AI experience, passionate about building intelligent systems and Agentic AI applications that solve real-world problems. Experienced in developing machine learning pipelines, Retrieval-Augmented Generation (RAG) systems, conversational AI agents, sentiment analysis solutions, LLM fine-tuning, and data engineering workflows. Skilled in Python, LLM integration, prompt engineering, containerised deployment with Docker, and modern AI application development.

EXPERIENCE

AI/ML Engineer Intern | *Strata Nextgen Pty Ltd | 6 months*

- Built an email-channelling system that receives new mail through Microsoft Graph API webhooks and classifies each message with an LLM.
- Orchestrated the processing pipeline with Temporal for durable, retry-safe workflows around LLM calls.
- Resolved Azure AD admin-consent requirements for Graph API access and set up Claude Code with CLAUDE.md and skill files for the team workflow.
- Worked on predicting 3D Gaussian Splatting quality and diagnosing capture failures from COLMAP metrics as proxy signals, avoiding full GPU training runs.

AI/ML Engineer Intern | *Sri Lanka Telecom (SLT-MOBITEL) | 6 months*

- Built Ask SLT, a multi-agent RAG assistant (FastAPI, LangGraph) in which a supervisor routes queries to specialist agents for HR, Finance, IT, and Legal.
- Rebuilt the caching layer as a semantic cache using Qdrant and Redis, calibrating similarity thresholds and fixing cache-poisoning and request-coalescing issues.
- Constructed a supervised fine-tuning dataset from internal HR policy documents in Alpaca and chat-message JSONL formats.
- Fine-tuned TinyLlama-1.1B with Full Fine-Tuning, LoRA, and QLoRA, evaluated with BERTScore, F1, and Exact Match, and published the LoRA and QLoRA models on Hugging Face.

PROJECTS

WareFlow AI: Warehouse Robotics Simulation | *LangGraph, FastAPI, Qdrant*

- Co-developed a three-agent warehouse robot fleet coordination simulation in a four-member team (IT3041 Agentic AI Systems).
- Owned the Task Planning Agent in LangGraph and evaluated it against 17 prompt-injection and jailbreak test cases.

Urban Flow Analytics: Taxi Fare Prediction | *Python, scikit-learn | CodefestDatathon2026*

- Built a taxi fare prediction pipeline covering preprocessing, anomaly detection, and feature engineering under an upfront-pricing constraint.

Domain-Specific LLM Fine-Tuning | *Llama 2 7B Chat, Hugging Face Transformers, PEFT, PyTorch*

- Fine-tuned the Llama 2 7B Chat model on a domain-specific HR question-answering dataset using Full Fine-Tuning, LoRA, and QLoRA approaches.
- Developed an end-to-end fine-tuning pipeline, including dataset preprocessing, instruction formatting, tokenization, and train/validation/test splitting.
- Implemented and compared Full Fine-Tuning, LoRA, and QLoRA based on validation loss, training efficiency, GPU memory utilization, and inference performance.
- Configured LoRA adapters using PEFT by targeting transformer attention and feed-forward projection layers (q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, and down_proj).
- Built inference pipelines for the fine-tuned models and published trained models through the Hugging Face ecosystem.
- Utilized PyTorch, Hugging Face Transformers, TRL, PEFT, and BitsAndBytes to implement parameter-efficient fine-tuning workflows.

AI Research Assistant Agent | *Python, Flask, OpenRouter, Tavily Search, SQLite, ReportLab*

- Developed an Agentic AI research assistant capable of answering complex research queries using LLM reasoning, web search, conversation memory, and automated report generation.
- Implemented tool-calling workflows with Tavily Search, enabling the agent to retrieve and synthesise real-time information from external sources.

- Built a persistent conversation memory system using SQLite to maintain context across multi-turn interactions and improve response relevance.
- Designed agent decision logic to dynamically determine when external web search was required versus direct LLM reasoning.
- Generated structured research reports with executive summaries, key findings, detailed analysis, future outlook, and PDF export functionality.
- Integrated OpenRouter-hosted LLMs (Qwen) with a Flask web application to deliver an end-to-end AI research workflow.

Enterprise RAG Document Assistant | *FastAPI, LangChain, ChromaDB, Groq (Llama 3.1), HuggingFace, Docker*

- Built a full-stack Retrieval-Augmented Generation (RAG) API enabling PDF upload and natural-language querying, powered by LangChain and Groq-hosted Llama 3.1 for low-latency inference.
- Engineered a semantic search pipeline using local HuggingFace embeddings (all-MiniLM-L6-v2) persisted in ChromaDB, with context-grounded prompting to prevent hallucinated answers.
- Designed a modular FastAPI backend with PDF chunking, recursive text splitting, and CORS-enabled endpoints for scalable document ingestion and chat-based retrieval.
- Containerised the application using Docker to ensure consistent, portable deployment across environments.

Sentiment Analysis System | *Python, Scikit-learn, Flask*

- Developed an end-to-end sentiment analysis pipeline using TF-IDF, Chi-Square feature selection, and Logistic Regression.
- Applied NLP preprocessing, feature engineering, and model evaluation techniques to classify movie reviews as positive or negative.
- Deployed the trained model through a Flask web application for real-time sentiment prediction.

Formula 1 Data Warehouse & BI Pipeline | *SSIS, SQL Server, SSAS, Power BI*

- Built a complete data engineering pipeline using an SSIS-driven ETL framework to ingest, clean, and transform semi-structured historical Formula 1 data.
- Designed a Star Schema Data Warehouse in SQL Server with Surrogate Keys and Slowly Changing Dimensions (SCD Type 2) to preserve historical accuracy without data redundancy.
- Developed an SSAS Multidimensional Cube as a semantic layer, powering interactive Power BI dashboards and Excel OLAP analysis for trend discovery.

Data Analysis & Predictive Modelling | *R*

- Designed and implemented multiple predictive modelling projects in R, applying linear regression, hypothesis testing, and advanced data visualisation to extract actionable insights.
- Applied statistical inference techniques to validate model assumptions and communicate findings through reproducible R Markdown reports.

Fish Market System | *MERN Stack*

- Built a Fish Market management platform using the MERN stack, featuring real-time bidding, transaction tracking, and a responsive React frontend.

Campus Management System | *Spring Boot*

- Developed a full-featured campus management system using Spring Boot, implementing RESTful APIs and backend business logic for student, course, and staff management.

SKILLS

AI / ML: Machine Learning, NLP, LLM Fine-Tuning, Full Fine-Tuning, LoRA, QLoRA, PEFT, Hugging Face Transformers, Llama 2 7B Chat, Sentiment Analysis, RAG, Agentic AI, LLM Applications, Prompt Engineering, Tool Calling, LangChain, HuggingFace Hub, Transformers (BERT), ChromaDB, Gemini API, Groq, OpenRouter, Tavily Search, LangGraph, Qdrant, Redis, Temporal, Semantic Caching, BERTScore

Data Science: Statistical Modelling, Regression, Hypothesis Testing, Data Visualisation, R, Python (Pandas, NumPy, Scikit-learn, Matplotlib)

Data Engineering: ETL/SSIS, Star Schema Design, SCD Type 2, SQL Server, SSAS, Power BI, Excel OLAP

Programming: Python, R, Java, C, C++

Databases: MySQL, Oracle Database, MongoDB

Web & Backend: Next.js, FastAPI, Flask, Spring Boot, MERN Stack, REST APIs

DevOps & Tools: Docker, Linux, Git, VirtualBox

EDUCATION

BSc (Hons) IT Specializing in Data Science, SLIIT	2023 – 2027
GCE A/L – Ananda College	2020(21)
GCE O/L – Ananda College	2017

CERTIFICATIONS

AI / ML Course

- Covered supervised and unsupervised learning algorithms including regression, classification, clustering, and model evaluation.

Python Programming

MLOps Course – KodeKloud (In Progress)

- Currently covering ML deployment, model monitoring, CI/CD pipelines for ML, and production-grade MLOps practices.

ACHIEVEMENTS

- Built a personal AI/data lab to prototype and experiment with LLM integrations, Agentic AI workflows, containerised deployments, and data pipelines.